Available online at www.sciencedirect.com
SCIENCE DIRECT®
ELSEVIER
Tectonophysics 362 (2003) 105–122
TECTONOPHYSICS
www.elsevier.com/locate/tecto

Paleomagnetism and geochronology of an Early Proterozoic quartz diorite in the southern Wind River Range, Wyoming, USA

Stephen S. Harlan[a,*], John W. Geissman[b], Wayne R. Premo[c]

[a] *Department of Environmental Science and Policy, George Mason University, Fairfax, VA 22030-4444, USA*
[b] *Department of Earth and Planetary Sciences, University of New Mexico, Albuquerque, NM 87131, USA*
[c] *US Geological Survey, P.O. Box 25046, MS 963, Denver Federal Center, Denver, CO 80225, USA*

Received 9 January 2001; received in revised form 26 June 2001; accepted 8 July 2001

Abstract

We present geochronologic and paleomagnetic data from a north-trending quartz diorite intrusion that cuts Archean metasedimentary and metaigneous rocks of the South Pass Greenstone Belt of the Wyoming craton. The quartz diorite was previously thought to be either Archean or Early Proterozoic (?) in age and is cut by north and northeast-trending Proterozoic diabase dikes of uncertain age, for which we also report paleomagnetic data. New U–Pb analyses of baddeleyite and zircon from the quartz diorite yield a concordia upper intercept age of 2170 ± 8 Ma (95% confidence). An $^{40}Ar/^{39}Ar$ amphibole date from the same sample yields a similar apparent age of about 2124 ± 30 Ma (2σ), thus confirming that the intrusion is Early Proterozoic in age and that it has probably not been thermally disturbed since emplacement. A magmatic event at ca. 2.17 Ga has not previously been documented in the Wyoming craton. The quartz diorite and one of the crosscutting diabase dikes yield essentially identical, well-defined characteristic remanent magnetizations. Results from eight sites in the quartz diorite yield an in situ mean direction of north declination and moderate to steep positive inclination (Dec. = 355°, Inc. = 65°, $k = 145$, $\alpha_{95} = 5°$) with a paleomagnetic pole at 84°N, 215°E ($\delta m = 6°$, $\delta p = 7°$). Data from other diabase dike sites are inconsistent with the quartz diorite results, but the importance of these results is uncertain because the age of the dikes is not well known. Interpretation of the quartz diorite remanent magnetization is problematic. The in situ direction is similar to expected directions for magnetizations of Late Cretaceous/early Tertiary age. However, there is no compelling evidence to suggest that these rocks were remagnetized during the late Mesozoic or Cenozoic. Assuming this magnetization to be primary, then the in situ paleomagnetic pole is strongly discordant with poles of 2167, 2214, and 2217 Ma from the Canadian Shield, and is consistent with proposed separation of the Wyoming Craton and Laurentia prior to about 1.8 Ga. Correcting the quartz diorite pole for the possible effects of Laramide-age tilting of the Wind River Range, based on the attitude of nearby overlying Cambrian Flathead Sandstone (dip = 20°, N20°E), gives a tilt corrected pole of 75°N, 58°E ($\delta m = 4°$, $\delta p = 6°$), which is also discordant with respect to time-equivalent poles from the Superior Province. Reconstruction of the Superior and Wyoming Province using a rotation similar to that proposed by Roscoe and Card [Can. J. Earth Sci. 46(1993)2475] is problematic, but reconstruction of the Superior and Wyoming Provinces based on restoring them to their correct paleolatitude and orientation using a closest approach fit indicates that the two cratons could have been adjacent at about 2.17 Ga prior to rifting at about 2.15 Ga. The paleomagnetic data presented are consistent with the hypothesis that the

* Corresponding author. Tel.: +1-703-993-3892; fax: +1-703-993-1216.
E-mail address: sharlan@gmu.edu (S.S. Harlan).

0040-1951/02/$ - see front matter © 2002 Elsevier Science B.V. All rights reserved.
doi:10.1016/S0040-1951(02)00633-9

Huronian and Snowy Pass Supergroups could have evolved as part of a single epicratonic sedimentary basin during the Early Proterozoic.
© 2002 Elsevier Science B.V. All rights reserved.

Keywords: Paleomagnetism; Geochronology; Diorite

1. Introduction

Recent interest in the existence of and continental configurations for the proposed Neoproterozoic supercontinent Rodinia (Dalziel, 1991; Moores, 1991) has spurred further research in the field of Proterozoic plate tectonics. Fundamental questions exist regarding the style, nature, and mechanisms of plate tectonic activity during the early Precambrian as well as possible Proterozoic plate configurations that may have predated Rodinia. Recently, Williams et al. (1991) and Roscoe and Card (1993) proposed the existence of "Kenorland" as a Late Archean/Early Proterozoic supercontinent composed of the Wyoming, Superior, and possibly other cratons (Fig. 1) that were assembled prior to about 2.45 Ga. Specific evidence in favor of this reconstruction is the stratigraphic and sedimentological similarities between the Lower Proterozoic Huronian Supergroup along the present-day southern margin of the Canadian Shield and the Snowy Pass Supergroup along the southern margin of the Wyoming Province. Roscoe and Card postulated that these stratigraphic sequences were deposited in a single, broad epicratonic basin that straddled the Archean basement of the Superior and Wyoming cratons. Subsequent rifting at about 2.2 to 2.15 Ga dismembered Kenorland and separated the Wyoming and Superior cratons (Fig. 1). Both the Wyoming and Superior cratons were subsequently reassembled at about 1.85 Ga into their present relative configurations. In this paper we present new paleomagnetic and geochronologic re-

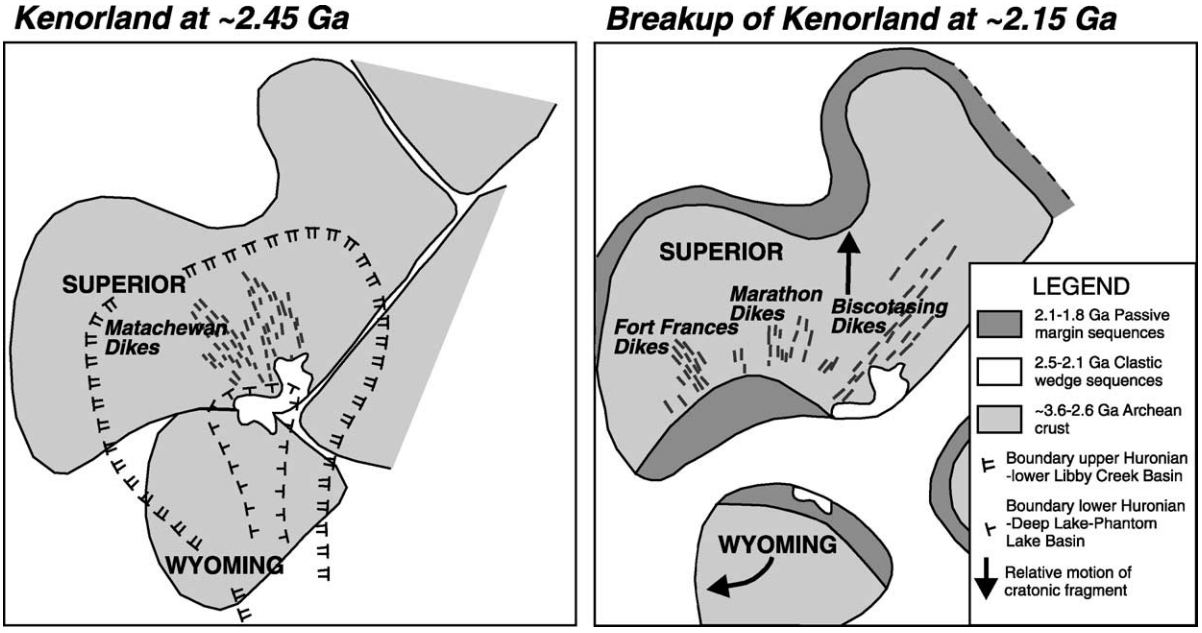


Fig. 1. Proposed model for the relative positions of the Wyoming and Superior cratons as part of an late Archean/early Paleoproterozoic supercontinent "Kenorland" and its breakup at about 2.20–2.15 Ga. Modified from Roscoe and Card (1993).

sults from a Proterozoic quartz diorite intrusion and spatially related diabase dikes exposed in the southern Wind River Range of Wyoming, and investigate the implications of these data for the fragmentation of Kenorland.

As part of an ongoing investigation of the age and paleomagnetism of Proterozoic dikes exposed in several foreland uplifts of the Wyoming Province, we sampled a quartz diorite intrusion and several diabase dikes in the southern part of the Archean South Pass Greenstone Belt (Fig. 2). Our goal is to provide well-dated and paleomagnetically well-defined poles to contribute to an apparent polar wander (APW) path for the Wyoming craton and to use these data to test competing models for the assembly of possible Precambrian plate configurations. In this study we examine the data from these rocks in the context of the proposed model for Kenorland of Roscoe and Card (1993).

2. Geologic setting

The Wyoming Province is one of seven Archean provinces defining Laurentia, the Precambrian core of the North American craton. Although Sm–Nd isotopic evidence systems indicate that most of the material in the Wyoming Province separated from the mantle in the Early and Middle Archean (Aleinikoff et al., 1989; Frost, 1993), most of the rocks in the province formed in the Late Archean. These rocks include migmatitic gneisses, late supracrustal successions, abundant granitic rocks, and lesser volumes of ultramafic to mafic intrusions. These are well exposed in several contractional uplifts that formed during the latest Cretaceous to early Tertiary Laramide Orogeny. The Wind River Range is dominated by high-grade metamorphic and metaigneous basement rocks (Frost and Frost, 1993; Frost et al., 2000). In the South Pass area, at the southern termination of the range (Fig. 2),

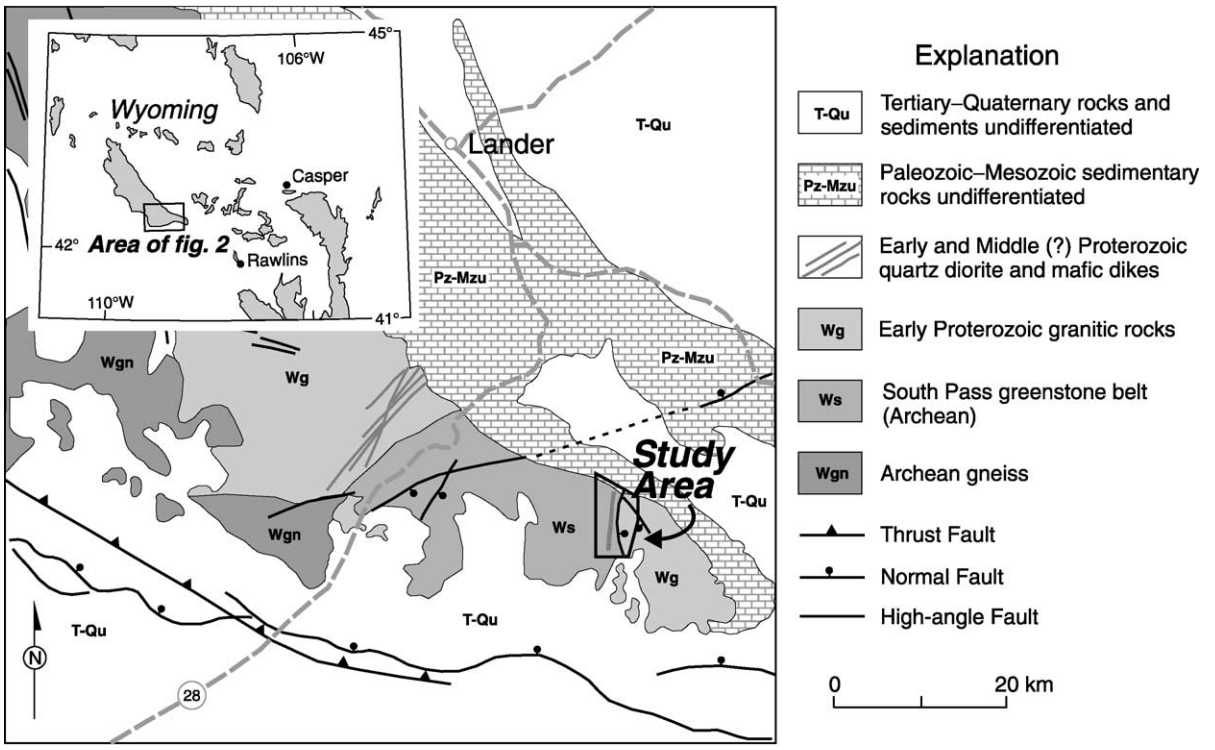


Fig. 2. Simplified geologic map of the southern Wind River Range, western Wyoming, showing the location of the paleomagnetic study. Modified from Love and Christiansen (1985).

a thick and lithologically complex supracrustal sequence termed the South Pass granite–greenstone belt (Bayley et al., 1973; Hausel, 1991) consists of regionally metamorphosed greenschist- to amphibolite-grade metasedimentary and mafic to ultramafic metaigneous rocks. These supracrustal rocks were subsequently juxtaposed against the Archean basement gneiss complex to the north along a northeast-trending shear zone and intruded by the ca. 2.63 Ga Louis Lake Batholith as well as smaller granitic plutons at about 2.565 Ga (Bayley et al., 1973; Stuckless et al., 1985; Hausel, 1991; Frost and Frost, 1993; Frost et al., 1998; Frost et al., 2000). All of these rocks are crosscut by a swarm of northeast-trending tholeiitic diabase dikes (Bayley et al., 1973; Hausel, 1991), ranging from 10 to 100 m in thickness, which have planar contacts and well-developed chilled margins. The age of these dikes is poorly known. Whole rock K–Ar dates reported by Condie et al. (1969) yield apparent ages of 2010 to 1270 Ma. K–Ar dates of 1880 to 1600 Ma for pyroxene separates from the dikes were reported by Spall

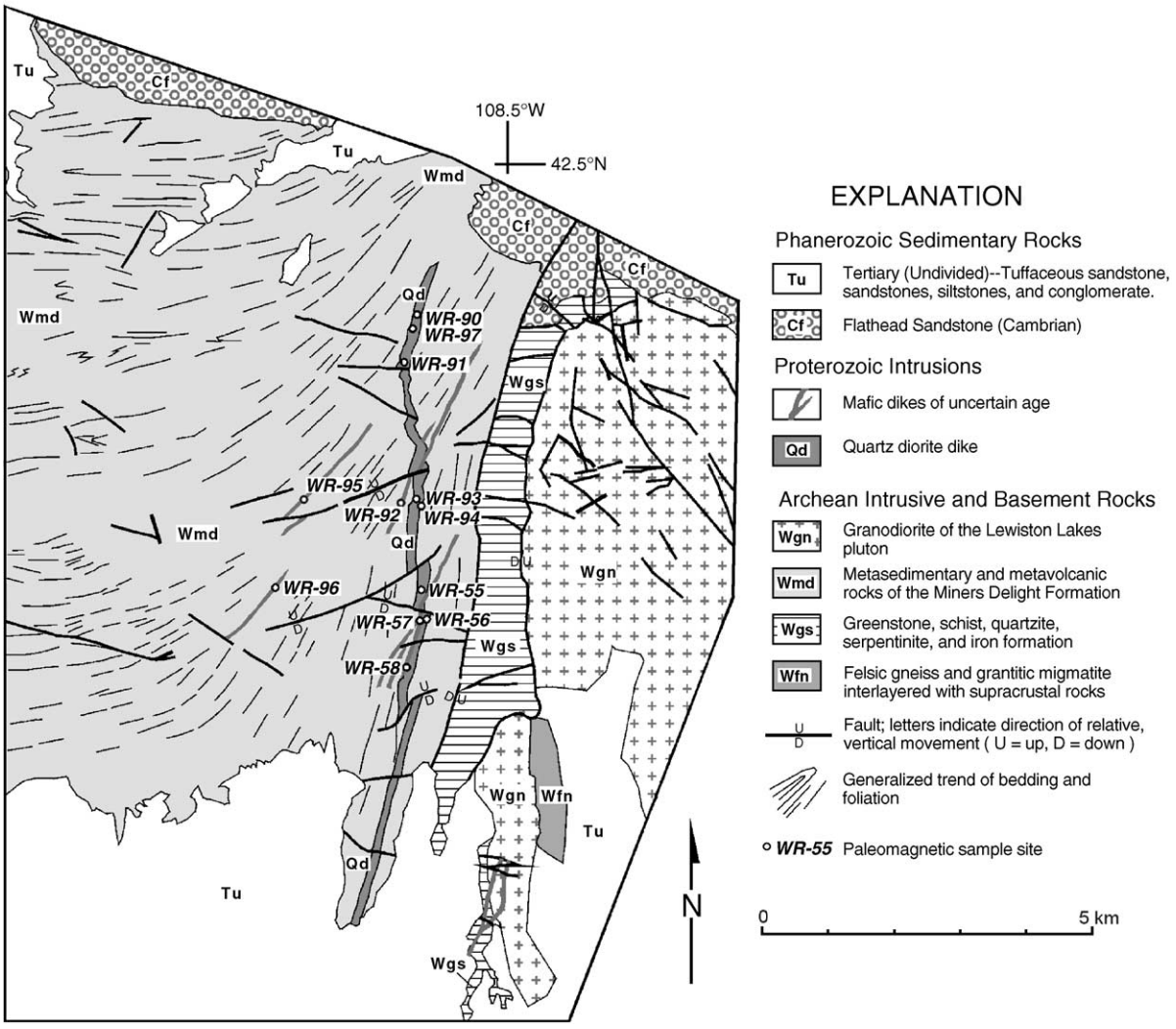


Fig. 3. Geologic map showing the location of paleomagnetic sampling sites in a north trending quartz diorite dike, as well as in four northeast-trending diabase dikes. Modified from Hausel (1991).

(1971). Because of problems associated with potential ^{40}Ar loss and the nearly ubiquitous presence of excess ^{40}Ar in low K phases such as pyroxene and plagioclase, both major constituents of the diabase dikes, the K–Ar data from the dikes cannot be considered reliable and hence the age of the dikes is poorly known. Attempts to obtain zircon and/or baddeleyite from the diabase dikes by us and other workers (K.R. Chamberlain, oral communication, 2000) have proven unsuccessful.

Southeast of Atlantic City, in the Lewiston Mining district (Radium Springs 7.5′ Quadrangle; Hausel, 1988), a unique quartz diorite intrusion, 11 km in north–south dimension, and up to 200 m in east–west dimension, cuts Archean metavolcanic and metasedimentary rocks of the Miners Delight Formation (Bayley et al., 1973; Hausel, 1988, 1991), a major component of the greenstone belt (Fig. 3). The quartz diorite is an unmetamorphosed medium-gray to slightly pink, fine- to medium-grained rock composed of plagioclase, pyroxene, amphibole, biotite, and quartz. Although the quartz diorite clearly post dates Archean lode gold mineralization in the Lewiston Mining district, Hausel (1988) interpreted the quartz diorite to be Archean in age, presumably because it is cut by northeast-trending diabase dikes (Fig. 3). However, at one locality, geologic mapping suggests that one of the diabase dikes is actually cut by the quartz diorite dike (cf. maps by Hausel, 1988, 1991). At another location, a diabase dike is exposed along the western margin of the quartz diorite. Field examination of this relationship shows no distinct chilled margin between the diabase and the quartz diorite, suggesting that they may have been emplaced at about the same time. A subsequent publication (Hausel, 1991) classified the quartz diorite as either Archean or questionable Early Proterozoic in age. Northeast and north of the sampling area, Precambrian rocks are nonconformably overlain by the Middle Cambrian Flathead Sandstone, which dips up to 20° to the northeast.

3. Methods

Samples for paleomagnetic study were collected using a portable field drill with individual samples oriented with magnetic and solar compasses and a clinometer. Samples from eight sites were collected from the quartz diorite intrusion and from four sites in three mafic dikes (Fig. 3). The contacts between the quartz diorite and host rocks are not well exposed, precluding any form of baked contact test with host sedimentary and metavolcanic rocks. Elsewhere, paleomagnetic results from the Archean metasedimentary rocks of the South Pass Greenstone Belt have failed to yield coherent paleomagnetic data at sites adjacent to dikes or well removed from the possible thermal effects of dikes (Harlan and Geissman, unpublished data).

The natural remanent magnetization (NRM) of each sample was measured using a superconducting rock magnetometer. Detailed progressive alternating field (AF), thermal, or AF followed by thermal demagnetization procedures were applied to all samples. Demagnetization results were analyzed using orthogonal vector diagrams, stereographic projections, and normalized intensity decay plots. Characteristic and secondary components of magnetization, recognized by collinear demagnetization paths on vector diagrams, were calculated using principal components analysis (Kirschvink, 1980). For samples in which demagnetization failed to isolate a single characteristic direction or stable endpoint but yielded a curvilinear demagnetization trajectory due to overlapping coercivity and/or unblocking temperatures, we calculated demagnetization circles (McFadden and McElhinny, 1988). Site-mean directions consisting of well-defined linear data were calculated using statistics from Fisher (1953); site-mean directions based on a combination of linear and great circle data were calculated using the maximum likelihood approach method of McFadden and McElhinny (1988). Identification of magnetic carriers was interpreted from the response of individual samples to AF and/or thermal demagnetization experiments.

For isotopic dating, a hand sample from paleomagnetic site WR55 weighing about 4 kg was crushed in a jaw crusher and disk mill. Zircon, baddeleyite, amphibole, and biotite from this sample were concentrated using a combination of standard gravimetric and magnetic separation techniques. The biotite showed evidence of chloritic alteration and was judged unsuitable for isotopic dating. Pure amphibole, zircon, and baddeleyite separates for dating were obtained by handpicking. Thin-section analysis indi-

cates that baddeleyite is associated with biotite, whereas zircon is associated with quartz. The presence of the two phases presumably reflects different silica activities during the crystallization of the rock. Prior to dissolution, individual zircon and baddeleyite fractions were cleaned very briefly in distilled 7 N HNO_3,, weighed into PFA teflon vials, and dissolved in distilled concentrated $HF + HNO_3$ in a large (6.5-cm diameter) Parr-type TFE teflon dissolution vessel at 210 °C for approximately 10–14 days using the HF-vapor technique of Krogh (1978). The fractions were then spiked with a $^{205}Pb-^{233}U-^{236}U-^{230}Th$ dilute tracer solution and reheated to achieve isotopic equilibration. U–Th–Pb were extracted from the dissolved fractions using AG 1-X8 anion exchange resin in a dilute HBr medium. Pb residues were dissolved in H_3PO_4 and loaded onto single Re filaments, and U–Th–Pb isotopic ratios were measured using a VG Isomass 54R mass spectrometer equipped with a Faraday cup and Daly multiplier. Pb isotopic ratios were corrected for mass fractionation of either 0.08 ± 0.03 % per a.m.u. (batch 83) or 0.12 ± 0.03 % per a.m.u. (batch 43); a laboratory blank varying between 31 pg total Pb with average composition of $^{206}Pb/^{204}Pb = 19.05 \pm 0.6$, $^{207}Pb/^{204}Pb = 15.65 \pm 0.2$, and $^{208}Pb/^{204}Pb = 38.65 \pm 0.75$, and 72 pg total Pb with average composition of $^{206}Pb/^{204}Pb = 19.42 \pm 0.77$, $^{207}Pb/^{204}Pb = 15.51 \pm 0.15$, and $^{208}Pb/^{204}Pb = 38.0 \pm 1.0$ (all errors at 2σ); and initial common Pb using values of Stacey and Kramers (1975) for an approximate age for the sample and second-stage $^{238}U/^{204}Pb = 9.74$.

U and Th were extracted using AG 1-X8 anion exchange resin in a 7 N HNO_3 medium and residues were loaded onto Re filaments using dilute HNO_3. U and Th ratios were measured in the triple filament mode, and corrected for a mass fractionation of $0.10 \pm 0.03\%$ per a.m.u. and laboratory blanks of 4 and 15 pg, respectively. Ages were calculated using decay constants from Steiger and Jäger (1977). Concordia intercept ages were determined using the algorithms of Ludwig (1980, 1985), which use the regression approach of York (1969); uncertainties in regressed ages are reported at the 95% confidence level.

$^{40}Ar/^{39}Ar$ analysis of the amphibole was conducted using facilities at the US Geological Survey, Denver, CO. The amphibole and neutron flux monitors were irradiated in the US Geological Survey TRIGA reactor. Neutron flux was monitored using MMhb-1, which has a K–Ar age of 520.4 ± 1.7 Ma (Samson and Alexander, 1987). Apparent ages were calculated using the decay constants of Steiger and Jäger (1977). A detailed description of analytical procedures similar to those used in this study is described in Table 1 of Harlan and Geissman (1998).

4. Results

4.1. Paleomagnetic results

The NRM intensity values for quartz diorite samples from the eight sites range from 6.37×10^{-4} to 3.25 A/m, with a geometric mean of 6.57×10^{-2} A/m. All eight sites in the quartz diorite yield a well-defined remanent magnetization of north declination and moderate positive inclination, as revealed by the linear decay of the remanence vector in most (87%) samples to the origin of demagnetization diagrams in alternating field, thermal, and low temperature followed by thermal demagnetization experiments (Fig. 4A–D). In a small percentage of samples (13%), the characteristic remanent magnetization could not be readily isolated during demagnetization experiments because of the overlap of unblocking temperature/coercivity spectra of randomly directed secondary magnetizations. Typically, these samples showed great circle trends in equal-area projections that trended toward the characteristic direction defined by the well behaved samples. Such samples were analyzed using the maximum likelihood approach method of McFadden and McElhinny (1988) in order to determine the site-mean directions. The combination of moderate to high coercivities in alternating field demagnetization, distributed to knee-shaped thermal demagnetization intensity curves, and maximum laboratory unblocking temperatures of 580 to 590 °C (Fig. 4C and D) indicate that the characteristic remanent magnetization is carried by magnetite. Site-mean directions are moderate- to well grouped with α_{95} values of 2° to 12° (Fig. 5A), with k values ranging from 16 to 136 (Table 1). The eight sites provide an in situ grand mean-direction of declination = 355°, inclina-

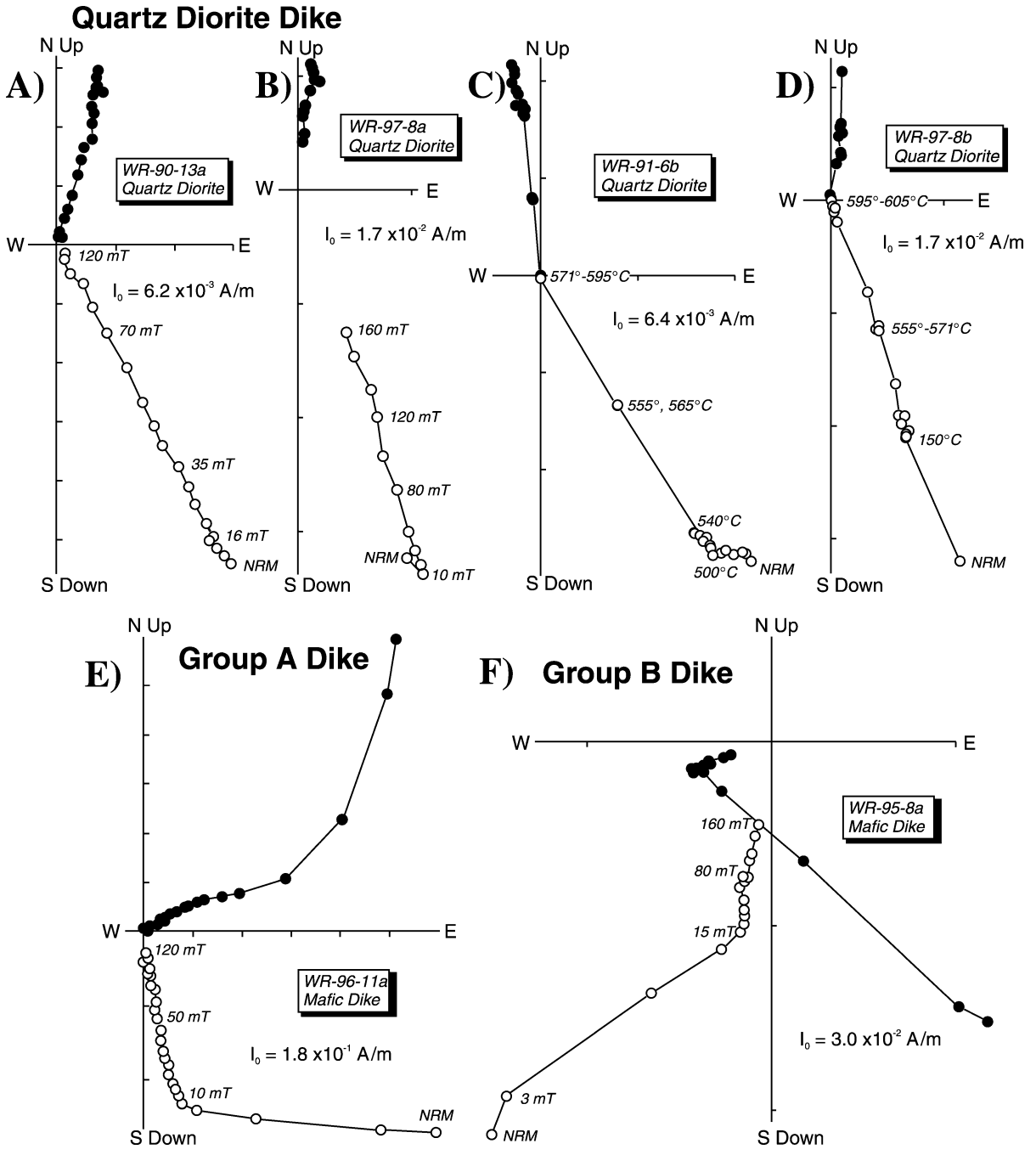


Fig. 4. Representative orthogonal vector diagrams from quartz diorite and diabase dike samples showing the declination (solid circles) and true inclination (open circles) of the remanent magnetization during progressive alternating field and thermal demagnetization. I_0 = intensity of the natural remanent magnetization prior to demagnetization. Quartz diorite samples in (B) and (D) are specimens from the same sample and have been subjected to alternating field and thermal demagnetization, respectively.

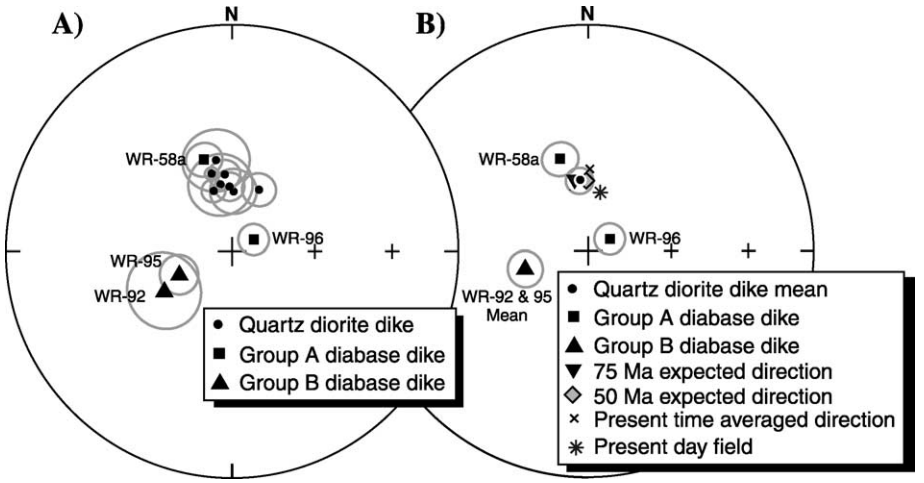


Fig. 5. (A) Equal-area plot showing the in situ site-mean magnetization directions from individual sites in the quartz diorite and diabase dikes and their α_{95} cones of confidence. (B) Equal-area projection showing the in situ group-mean direction from the quartz diorite and the mean directions from three diabase dikes. Also shown are the expected directions for the Late Cretaceous and early Tertiary, as well as the present-day and time-averaged fields. Only the site- and group-mean directions for the diabase dikes are labeled.

tion = 65°, $k = 145$, and $\alpha_{95} = 5°$ (Table 1; Fig. 5B), with a corresponding paleomagnetic pole at 84°N, 215°E ($\delta m = 6°$, $\delta p = 7°$).

The NRM intensity values for most of the samples from the four diabase dikes sampled near the quartz diorite are significantly greater than the quartz diorite,

Table 1
Paleomagnetic site- and group-mean results from the quartz diorite intrusion and diabase dikes in the southern Wind River Mountains, Wyoming

Site	Slat.	Slong.	N/N_o	Dec.	Inc.	k	α_{95}	Plat.	Plong.
Quartz Diorite									
WR-55	42.4241°N	108.4241°W	12(4)/12	350.0°	65.6°	16	11.3°	81.2°N	202.3°E
WR-56	42.4220°N	108.5118°W	6/7	349.9°	56.4°	26	11.6°	80.5°N	129.7°E
WR-58b	42.4155°N	108.5146°W	11/11	345.1°	61.0°	39	2.3°	79.0°N	164.6°E
WR-90	42.4674°N	108.5264°W	12/12	23.6°	65.8°	53	6.0°	72.6°N	314.7°E
WR-91	42.4597°N	108.5184°W	10(3)/10	1.1°	68.4°	35	8.2°	80.8°N	255.8°E
WR-93	42.4381°N	108.5251°W	10/10	342.8°	67.2°	136	4.2°	76.0°N	199.7°E
WR-94	42.4366°N	108.5251°W	8(3)/9	354.2°	62.2°	107	5.5°	85.6°N	177.3°E
WR-97	42.4701°N	108.5179°W	10/10	357.7°	66.7°	88	5.2°	83.0°N	239.0°E
Mean			8	354.9°	64.6°	145	4.6°		
Mafic Dikes									
Group A Dikes (~ 2.1 Ga?)									
WR-58a	42.4155°N	108.5146°W	9(3)/10	342.9°	55.2°	72	6.3°	75.2°N	140.4°E
WR-96	42.4362°N	108.5441°W	11/11	61.0°	81.2°	70	5.5°	48.6°N	274.5°E
Group B Dikes (~ 1.4 Ga?)									
WR-92	42.4377°N	108.5217°W	9(5)/11	240.2°	61.4°	17	13.5°	10.7°N	210.9°E
WR-95	42.4399°N	108.5443°W	14/18	256.0°	69.1°	33	7.0°	20.7°N	215.1°E
Mean			23(5)/29	253.9°	66.6°	117	6.4°		

'Slat.' and 'Slong.' are the site latitude and longitude, respectively; N/N_o is the ratio of the number of samples used in the calculation of the site-mean directions to those demagnetized (the number in parentheses refers to the number of great circles used in calculation of the site-mean directions); 'Dec.' and 'Inc.' refer to the declination and inclination of the site-mean direction; k is precision parameter of Fisher (1953); α_{95} is the semi-angle of the cone of 95% confidence about the site-mean direction; and 'Plat.' and 'Plong.' are the latitude and longitude of the virtual geomagnetic pole calculated from the site-mean direction.

ranging from 5.4×10^{-3} to 17.1 A/m, with a geometric mean of 2.7×10^{-1} A/m. During AF and thermal demagnetization most dike samples yield well-defined remanent magnetizations of moderate to steep inclination (Fig. 4E and F). Site-mean directions are also moderate to well defined with α_{95} values ranging from 6° to 14° (Fig. 5A; Table 1). The dike that crosscuts the quartz diorite (WR58a) (Fig. 3) yields a magnetization (Dec. = 343°, Inc. = 55°, $\alpha_{95} = 6°$; Table 1) that is essentially identical to that of the host quartz diorite (WR-58b) (Fig. 5A and B). This paleomagnetic site was originally collected in order to conduct a paleomagnetic baked contact test for the diabase dikes against the host quartz diorite. As noted earlier, the contact of this dike is not sharply defined, but is gradational with the quartz diorite. These results suggest that the magnetization in the dike and the quartz diorite may have been acquired at about the same time. Because of the similarity of paleomagnetic directions and the possibility that its remanent magnetization may be similar in age to that of the quartz diorite, the magnetization is termed paleomagnetic group A. Two other dikes (WR-92 and WR-95) yield essentially identical magnetizations of west–southwest declination and moderate positive inclination that are termed paleomagnetic group B dikes. A mean direction based on 23 pooled sample directions from the two group B dikes that yield statistically identical mean directions gives a declination of 254°, inclination of 67°, $k = 117$, and $\alpha_{95} = 6°$ (Table 1; Fig. 5B), with a virtual geomagnetic pole at 22°N, 209°E ($\delta m = 8°$, $\delta p = 10°$. A fourth dike (site WR-96, Table 1; Fig. 5A and B) yields a mean magnetization of east–northeast declination and steep positive inclination (Dec. = 61°, Inc. = 81°, $\alpha_{95} = 6°$). The paleomagnetic direction of this dike is distinct from either group A or group B. Because of its relatively steep inclination, we tentatively assign this site to paleomagnetic group A, although it is possible that this could represent a third distinct paleomagnetic direction.

4.2. Geochronology

4.2.1. U–Pb analyses

Six baddeleyite fractions and one zircon fraction were analyzed from quartz diorite sample, WR-55. Baddeleyite grains were typically broken, brown to gray to tan, translucent to opaque flakes 20 to 70 μm in length, showing lengthwise cleavages and/or striations (Fig. 6B, inset). Many grains show a characteristic tongue and groove structure similar to those reported from the Mackenzie dike and sill samples from Canada (see Fig. 2c and d of Heaman and LeCheminant, 1993). In the Wyoming Province, similar baddeleyite grains have been observed in more mafic bodies of the pre-Stillwater Complex (Premo et al., 1990) and from a metagabbro plug that crosscuts the Deep Lake Group in the Sierra Madre of Wyoming, south of the Wind River Range (Premo and Van Schmus, 1989). Zircon, in contrast to baddeleyite, exhibited very discrete, euhedral, translucent to opaque, stubby dipyramidal prisms, with length-to-width ratios of about 1.5 to 1.

U and Th concentrations in baddeleyite are uniform between 500 and 650 ppm (Table 2), typical of intermediate-composition igneous rocks. However, unusually high U and Th concentrations (1900 and 3300 ppm, respectively) were found in zircons, but such high concentrations relative to baddeleyite are typical of primary zircons in mafic intrusions (Krogh et al., 1987; Heaman et al., 1992; Lanyon et al., 1993; Romer et al., 1995; Wingate and Giddings, 2000; Hanley and Wingate, 2000). A concordia plot of the U–Pb results (Fig. 6A) reveals that these fractions have undergone substantial Pb loss (between 20% and 30% for baddeleyite and ~ 80% for zircon). The degree of discordance within these analyses corresponds to U content; i.e., the greater the U content, the more discordant the analysis. The discordance shown by the baddeleyite grains contrasts with U–Pb data from baddeleyite that typically show discordance of no more than a few percent. Despite Pb loss, the quasi-linear array of baddeleyite and zircon analyses yielded a U–Pb concordia upper-intercept age of 2170 ± 8 Ma (95% confidence level) with a lower-intercept age of 116 ± 26 Ma (Fig. 6A). An alternative calculation, excluding the single zircon analysis because of its high degree of discordance, yields a statistically indistinguishable, although analytically less precise upper intercept age of 2166 ± 16 Ma (MSWD = 33) (Fig. 6A) and a lower intercept of 94 ± 69 Ma (2σ). Because baddeleyite is rarely found as a xenocrystic phase (Heaman and LeCheminant, 1993), we consider the baddeleyite present in the quartz diorite to be a primary mineral phase that

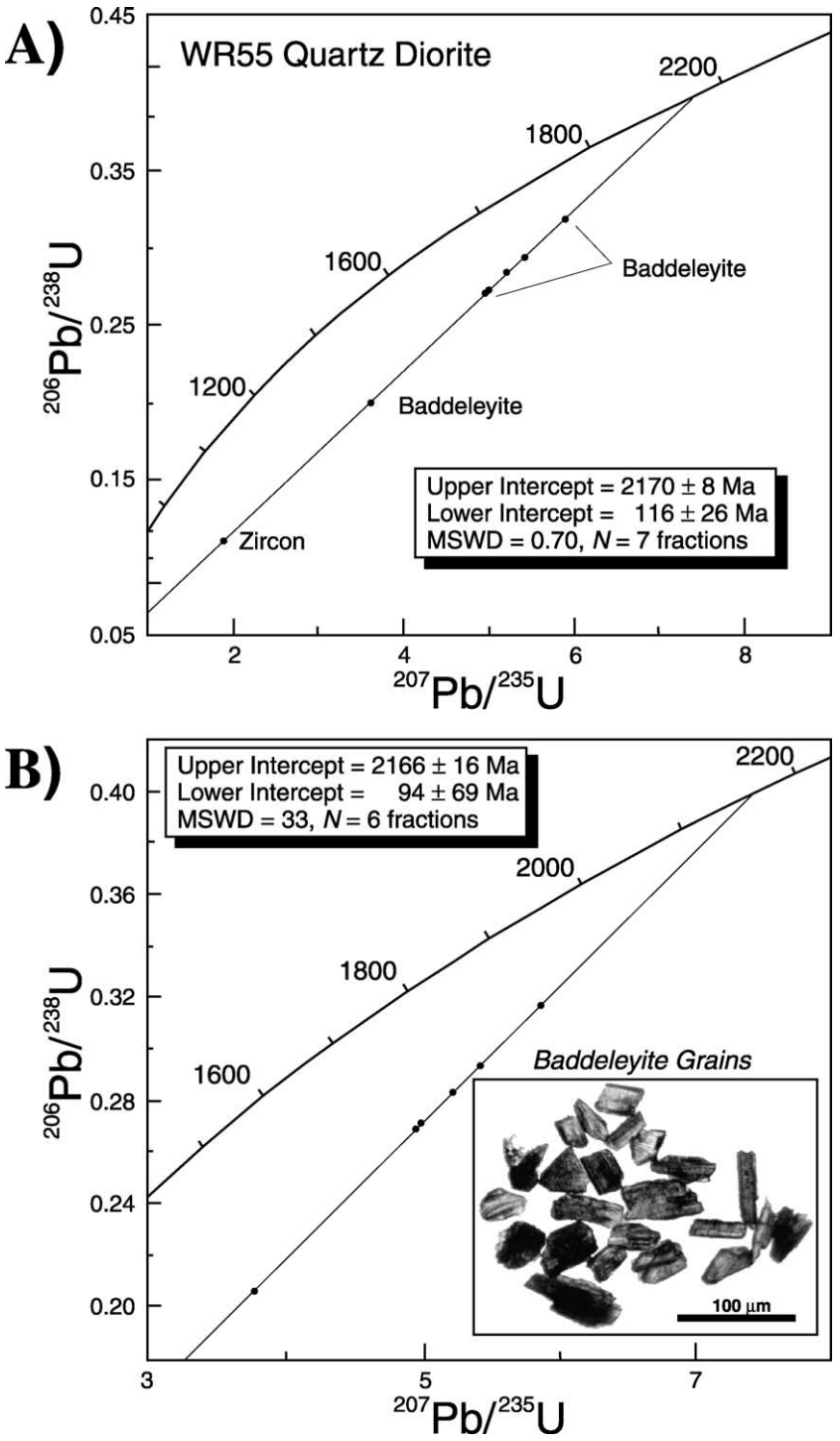

A)
WR55 Quartz Diorite
2200
1800
1600
1200
Baddeleyite
Baddeleyite
Zircon
Upper Intercept = 2170 ± 8 Ma
Lower Intercept = 116 ± 26 Ma
MSWD = 0.70, N = 7 fractions
0.45
0.35
0.25
0.15
0.05
2
4
6
8
206Pb/238U
207Pb/235U
B)
Upper Intercept = 2166 ± 16 Ma
Lower Intercept = 94 ± 69 Ma
MSWD = 33, N = 6 fractions
2200
2000
1800
1600
Baddeleyite Grains
100 μm
0.40
0.36
0.32
0.28
0.24
0.20
3
5
7
206Pb/238U
207Pb/235U

Table 2
U–Pb analytical data for baddeleyite and zircon from sample WR-55 from the quartz diorite intrusion, southern Wind River Mountains, Wyoming

Sample/fraction	Fraction specifics	Sample weight (mg)	U (ppm)	Th (ppm)	Pb (ppm)	$^{206}Pb/^{204}Pb$[a]	$^{207}Pb/^{206}Pb$[b]	$^{208}Pb/^{206}Pb$[b]	$^{206}Pb/^{238}U$[c]	$^{207}Pb/^{235}U$[c]	$^{207}Pb/^{206}Pb$[c]	$^{207}Pb/^{206}Pb$ age (Ma)[d]
WR-55 3(43)	Bd, hp	0.084	597	531	230	1843.7 (0.078)	0.14105 (0.053)	0.256161 (0.085)	0.31749 (0.118)	5.8872 (0.123)	0.13449 (0.036)	2157 (0.62)
WR-55 4(43)	Bd, hp	0.209	631	660	234	1170.3 (0.566)	0.14509 (0.035)	0.308728 (0.064)	0.29286 (0.180)	5.4224 (0.190)	0.13429 (0.058)	2155 (1.0)
WR-55 1(83)	Bd, 1 grain	0.004	938	1061	273	243.38 (0.27)	0.16930 (1.57)	0.39516 (1.65)	0.20610 (0.679)	3.7623 (0.703)	0.13240 (0.183)	2130 (3.2)
WR-55 2(83)	Zr, hp	0.033	1911	3294	344	225.91 (0.37)	0.18165 (0.163)	0.57231 (0.127)	0.11131 (0.260)	1.9166 (0.345)	0.12489 (0.060)	2027 (3.6)
WR-55 3(83)	Bd, hp	0.125	637	583	216	929.70 (1.00)	0.14644 (0.082)	0.28161 (0.112)	0.27182 (0.278)	4.9845 (0.308)	0.13300 (0.119)	2138 (2.1)
WR-55 4(83)	Bd, hp	0.174	555	916	193	612.10 (0.75)	0.15419 (0.068)	0.30193 (0.097)	0.26985 (0.334)	4.9602 (0.366)	0.13331 (0.134)	2142 (2.3)
WR-55 5(83)	Bd, hp	0.344	638	636	232	764.80 (0.22)	0.15050 (0.038)	0.30934 (0.068)	0.28356 (0.296)	5.2231 (0.300)	0.13359 (0.046)	2146 (0.8)

Numbers in parentheses are errors (in percent) at the 2-sigma level for the numbers directly above.
Bd = baddeleyite; Zr = zircon; and hp = handpicked.

[a] Corrected for fractionation only; batch #1: 0.12 ± 0.03% per a.m.u., and batch #2: 0.08 ± 0.04% per a.m.u.

[b] Corrected for fractionation and laboratory blank Pb; batch #1: 31 pg total Pb with a measured composition of $^{206}Pb/^{204}Pb = 19.05 \pm 0.6$, $^{207}Pb/^{204}Pb = 15.65 \pm 0.2$, and $^{208}Pb/^{204}Pb = 38.65 \pm 0.75$; batch #2: 75 pg total Pb with $^{206}Pb/^{204}Pb = 19.42 \pm 0.77$, $^{207}Pb/^{204}Pb = 15.51 \pm 0.15$, and $^{208}Pb/^{204}Pb = 38.0 \pm 1.00$.

[c] Radiogenic ratios; corrected for fractionation, laboratory blank Pb, and initial common Pb using values of Stacey and Kramers (1975) for an approximate age for the sample and second-stage $^{238}U/^{204}Pb = 9.74$.

[d] Age in millions of years, calculated using Ludwig (1980, 1985), and decay constants of Steiger and Jager (1977).

corresponds to the age of dike emplacement. However, there is no compelling reason to favor either of these age estimates and we choose to accept the combined zircon–baddeleyite age of 2170 Ma (MSWD = 0.7). We suggest that this age records Early Proterozoic emplacement of the quartz diorite. The lower-intercept age probably reflects uplift and dilatancy of the quartz diorite during the Late Cretaceous Laramide Orogeny. This Early Proterozoic age has not been previously recognized in the Wyoming Province and is thus unique to the magmatic history of this craton.

4.2.2. $^{40}Ar/^{39}Ar$ geochronology

Amphibole from the quartz diorite yields a total gas age of 2034 Ma and a somewhat complicated age spectrum (Fig. 7). Low-temperature steps (Table 3) show apparent argon loss but relatively high $^{39}Ar/^{37}Ar$ ratios, suggesting that argon is held in loosely bound interstitial sites. Three intermediate-temperature steps (1025 to 1075 °C) yield essentially identical apparent ages, relatively constant $^{39}Ar/^{37}Ar$ ratios characteristic of a uniform amphibole phase, and a plateau age of 2124 ± 30 Ma (2σ). Higher-temperature steps yield variable ages and low $^{39}Ar/^{37}Ar$ ratios (Table 3) that probably correspond to degassing of low potassium phases exsolved within the amphibole.

Despite the complications in the age spectrum, we interpret the plateau age of 2124 Ma to represent closure to ^{40}Ar diffusion. The amphibole age is somewhat younger than that from the zircon and baddeleyite, but the errors of the hornblende and baddeleyite dates overlap. We do not interpret the difference in

Fig. 6. (A) Concordia–discordia diagram showing the results of U–Pb analysis of six fractions of baddeleyite and one zircon fraction from the quartz diorite intrusion and the best-fit regression line. (B) Concordia–discordia diagram excluding the single zircon analysis. Inset shows a photograph of one of the baddeleyite fractions dated in this study. In both diagrams, the 2σ error ellipses associated with the individual analyses are smaller than the individual points. Uncertainties in ages are given at the 95% confidence level.

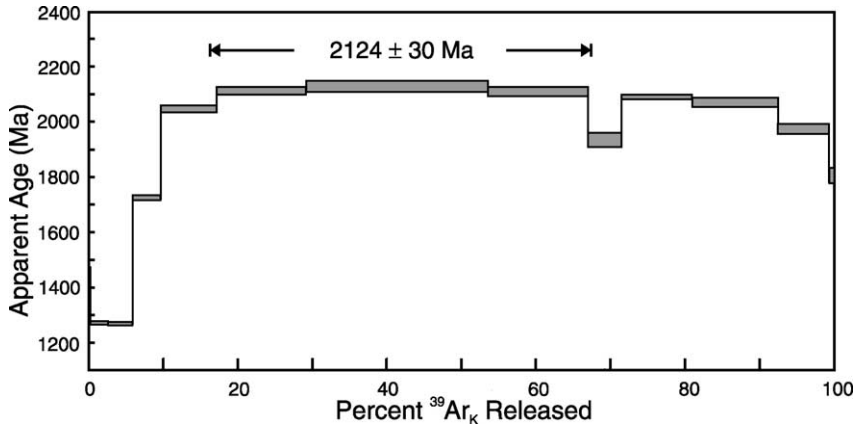


Fig. 7. $^{40}Ar/^{39}Ar$ age spectra for amphibole from the quartz diorite intrusion. The height of the rectangle for each temperature step represents the analytical error in the apparent age at $\pm 1\sigma$. The error in the plateaus date is reported at $\pm 2\sigma$.

apparent ages recorded by the two systems to reflect relatively slow cooling of the quartz diorite; rather this difference may be due to problems in uncertainty of the absolute age of the flux monitor used in this study (Renne et al., 1994; Min et al., 1999), uncertainties associated with decay constants for the K–Ar and U/Pb systems (Min et al., 1999; Begemann et al., 2001) that have not been factored into the U–Pb and $^{40}Ar/^{39}Ar$ dates, and observed complications in both the U–Pb and argon data. We do interpret the similarity in dates to reflect a relatively simple cooling history with no evidence of major thermal disturbance since emplacement and closure to ^{40}Ar diffusion in the Early Proterozoic.

Table 3
$^{40}Ar/^{39}Ar$ incremental heating data for amphibole from the quartz diorite intrusion, southern Wind River Mountains, Wyoming

Temperature (C)	$^{40}Ar_R$	$^{39}Ar_K$	$^{40}Ar_R/^{39}Ar_K$	$^{39}Ar/^{37}Ar$	$\%^{40}Ar_R$	$\%^{39}Ar$	Apparent age (Ma at 1σ)
Amphibole, 58.9 mg; measured $^{40}Ar/^{36}Ar_a = 298.9$; J-value = 0.014990 ± 0.2% (1σ)							
700	0.11660	0.00141	82.698	0.09	39.7	0.2	1455 ± 23
800	1.4675	0.02146	68.387	0.30	76.7	2.4	1273 ± 6
900	1.9650	0.02883	68.166	0.36	94.5	3.3	1270 ± 4
950	3.4545	0.03313	107.00	0.07	98.4	3.7	1726 ± 8
1000	9.3859	0.06659	140.96	0.12	98.4	7.5	2049 ± 11
1025	15.762	0.10605	148.64	0.16	99.7	12.0	2114 ± 15
1050	32.649	0.21670	150.69	0.18	99.7	24.4	2131 ± 19
1075	17.599	0.11866	148.31	0.18	99.8	13.4	2111 ± 15
1100	5.0965	0.03969	128.39	0.15	99.3	4.5	1936 ± 26
1125	12.236	0.08368	146.22	0.14	99.6	9.4	2094 ± 8
1150	14.718	0.10229	143.89	0.11	99.8	11.5	2074 ± 17
1175	7.9240	0.05959	132.98	0.06	99.4	6.7	1978 ± 16
1200	0.82420	0.00715	115.31	0.06	99.8	0.8	1811 ± 26
1250	0.19204	0.00148	130.09	0.01	94.7	0.2	1952 ± 50
Total gas			139.27				2034 ± 15

$^{40}Ar_R$ is radiogenic ^{40}Ar in volts signal; $^{39}Ar_K$ is potassium-derived ^{39}Ar in volts signal; $^{40}Ar_R/^{39}Ar_K$ is the ratio of $^{40}Ar_R$ to $^{39}Ar_K$ after correction for mass-discrimination and interfering isotopes; $^{39}Ar/^{37}Ar$ = ratio of $^{39}Ar_K$ to $^{37}Ar_{Ca}$ (this value can be converted to the approximate K/Ca by multiplying by 0.52); $\%^{40}Ar_R$ and $\%^{39}Ar$ are the percent of radiogenic ^{40}Ar and percent of total ^{39}Ar released in each temperature step. Temperature steps in boldface are those used in the calculation of the plateau age. Conversion of volts signal to moles Ar can be made using a conversion factor of 1.252×10^{-12} mol argon per volt of signal.

5. Discussion

The quartz diorite mean paleomagnetic direction based on eight well-distributed sites is strikingly similar to expected directions for the Late Cretaceous/Tertiary as well as present-day magnetic fields (Fig. 5B). For several reasons, however, we argue that the characteristic magnetization of the quartz diorite is a thermoremanent magnetization of Early Proterozoic age. First, the similarity between the U–Pb and the $^{40}Ar/^{39}Ar$ age determinations suggests the quartz diorite probably experienced a simple thermal history characterized by relatively rapid cooling and absence of any substantial thermochemical activity since the Early Proterozoic. Second, there is no geologic evidence for obvious Cretaceous/Tertiary thermochemical activity in the southern Wind River Mountains. Third, the maximum thickness of Phanerozoic strata covering this area, including through Maastrichtian time, when most motion along the Wind River thrust occurred, is estimated to be about 4.3 km (Royse, 1993). Even given a substantially higher geothermal gradient than today, such depths are probably insufficient to have caused complete thermoviscous remagnetization of the quartz diorite given the high laboratory unblocking temperatures of the characteristic remanent magnetization. Fourth, any Cretaceous/Tertiary chemical or thermal events would have had to preferentially remagnetize the quartz diorite and some of the diabase dikes, while leaving other nearby dikes apparently unaffected by such processes. Preferential remagnetization may be possible, but is considered unlikely. Finally, demagnetization behavior of the quartz diorite is typified by high median destructive fields and by high, narrow laboratory unblocking temperatures with complete demagnetization at temperatures below about 585 °C. Such behavior is indicative of fine (single-domain and pseudo-single domain) low-titanium magnetite particles as the dominant remanence carrier. Assemblages of such grains have been well documented as capable of carrying a magnetization of considerable antiquity.

Paleomagnetic results from diabase dikes in close proximity to the quartz diorite also yield well-defined remanent magnetizations, but the between site dispersion is considerably higher. One site (WR-58a) yields a magnetization identical to that of the quartz diorite and one yields a site-mean direction of steep inclination (WR-96); together these are tentatively referred to as paleomagnetic group A dikes, although it possible that the remanent magnetizations of the two dikes may not necessarily be related. The other two dikes yield southwest declinations and moderate to steep inclinations and are referred to as paleomagnetic group B dikes. The age(s) of the mafic dikes are not well known, but may range from about 2.1 Ga (i.e., slightly younger than the quartz diorite, group A?) to about 1470 Ma (group B?), based in part on recent geochronologic data from diabase dikes elsewhere in the Wind River Range and in the nearby Granite Mountains to the southeast (Chamberlain and Frost, 1995; K.R. Chamberlain, oral communication, 2001). It is possible that the swarm of northeast-trending diabase dikes in the southern Wind River Range may represent parallel dike swarms emplaced during more than one widely separated and discrete magmatic events. The emplacement of parallel to subparallel Proterozoic dike swarms of widely separated ages has been noted elsewhere in the Wyoming Province (Harlan et al., 1996; Harlan et al., 1997).

Application of field structural corrections to the paleomagnetic data from these Proterozoic rocks is somewhat problematic. We assume that the area has not been affected by any deformation of Precambrian age after intrusion of the quartz diorite or the diabase dikes, as there is no geologic evidence for such deformation and the margins of the quartz diorite and dikes are essentially vertical. Phanerozoic deformation in this area involved the Late Cretaceous/early Tertiary Laramide uplift of the Wind River Range and so the quartz diorite and mafic dikes may have experienced some, relatively recent tilt and/or vertical axis rotation. Plausible tilt corrections to the paleomagnetic data were applied based on the attitude of nearby exposures of Cambrian Flathead Sandstone (value of dip = 10 to 20°, dip direction N20°E). A 20° tilt correction shifts the quartz diorite site-mean direction to Dec. = 4°, Inc. = 46° with a paleomagnetic pole at 74.5°N, 58.1°E ($\delta p = 4°$, $\delta m = 6°$). Another possible source of tilt may include a slight (< 10°), southeast overall plunge of the southern Wind River Range. Correction for this plunge is problematic as any possible tilt corrections are poorly constrained. Hence,

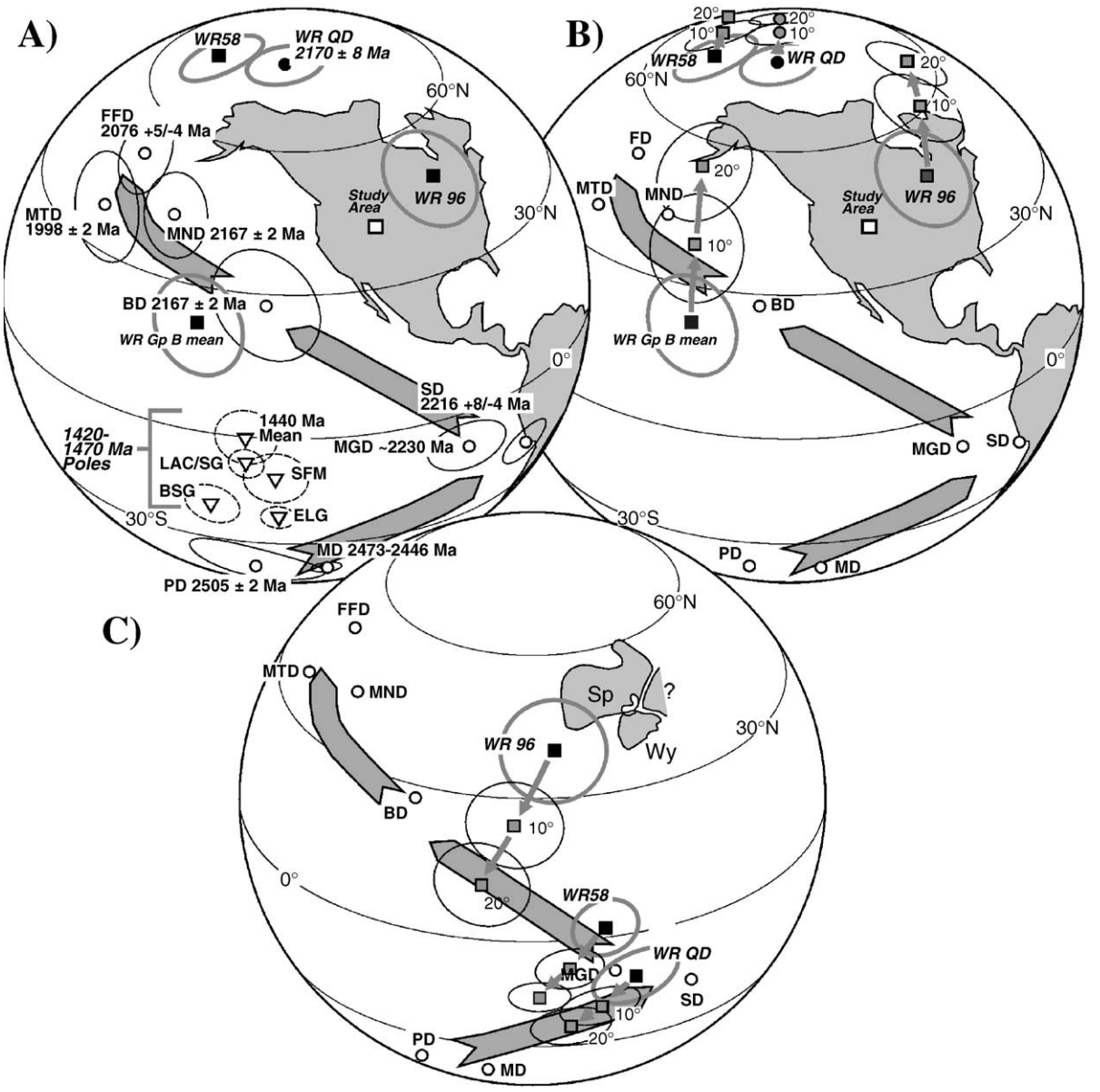
although we acknowledge that this may result in slightly greater uncertainties in pole positions, we have not taken this possible source of tilt into account. Finally, we consider the possibility that some local vertical axis rotation associated with uplift of the Wind River Range along the southwest-dipping Wind River thrust may have affected the Precambrian basement rocks in the hanging wall of the thrust. However, paleomagnetic results from Lower Triassic strata of the Chugwater Group in Red Canyon, about 30 km from the study area, yield a paleomagnetic pole that is in excellent agreement with other Chugwater Group and Moenkopi Formation poles from elsewhere in the Rocky Mountains (Van der Voo and Grubbs, 1977). This result suggests that localized vertical axis rotations associated with the Laramide uplift of the Wind River Range are minimal and could not have significantly affected the paleomagnetic results from the Proterozoic rocks.

In situ as well as structurally corrected paleomagnetic poles derived from data from the quartz diorite and two of the diabase dikes are discordant with respect to 2.4–2.0 Ga parts of the Early Proterozoic APW path for the Superior Province (Fig. 8A and B). The other two dikes (WR-92 and WR-95; group B mean) yield virtual geomagnetic poles that are similar to that provided by the 2167 Ma Biscotasing dikes of the Superior Province, but because the age of the Wind River diabase dikes is unknown, the importance of this comparison is not clear. The in situ group B mean direction pole lies 25° to 30° north and slightly west of most 1.5 to 1.4 Ga poles from North America. Given the likelihood that the group B mean does not adequately average paleosecular variation, it is permissible that the group B dikes record a magnetization of about 1470 Ma, consistent with isotopic age determinations of mafic dikes in the Granite Range to the east and dikes elsewhere in the Wind River Range (K.R. Chamberlain, oral communication).

Possible tilt corrections based on the orientation of nearby Cambrian strata move the quartz diorite and dike poles away from the APW path, increasing the sense of discordance (Fig. 8B). Further, if the group B dikes do carry a 1.47 Ga remanent magnetization, then the tilt corrections described above also move this pole farther away from other 1.5 to 1.4 Ga poles. This observation may imply that tilt corrections to our data based on the attitude of the Cambrian Flathead Sandstone are inappropriate, because the Precambrian basement rocks may have experienced less tilting than the sedimentary strata to the east.

Testing the Kenorland configuration of Roscoe and Card (1993) is problematic because they did not provide an Euler pole of rotation for their reconstruction. We provisionally test their reconstruction by calculating an Euler pole that restores the Wyoming Province and Superior cratons to a geometry similar to that shown in Fig. 1, located at about

Fig. 8. Orthographic global projections centered on 30°N, 225°E showing the present-day configuration and location of North, Central, and South America. (A) In situ virtual geomagnetic poles and 95% confidence ellipses from the quartz diorite (solid circle) and three mafic dikes (filled squares) in the southern Wind River Range, Wyoming. Also shown are well-defined and dated Early Proterozoic paleomagnetic poles (open circles) from the Superior craton that comprise the apparent polar wander path shown by the heavy gray arrows and selected 1470 to 1420 Ma poles from Laurentia (open triangles). (B) To investigate the possible affects of Late Cretaceous–early Tertiary deformation, tilt corrections were applied at 10° and 20° for the range of dips of Cambrian sandstone near the study area. The tilt corrections move all the Wind River Range virtual geomagnetic poles away from the Paleoproterozic apparent polar wander path for the Superior craton. Similarly, the group B dike pole of possible 1467 Ma age also moves away from ca. 1470–1420 Ma poles from Laurentia. (C) Orthographic projection centered on 30°N, 250°E showing reconstruction of the southern margin of the Superior craton with the Wyoming province following Roscoe and Card (1993) after a rotation of 172.6° about an Euler pole at 44.3°N, 264.8°E. Also shown are the in situ and 10° and 20° tilt corrected poles for the quartz diorite and sites WR58 and WR96. Assuming the quartz diorite represents a primary remanence at 2167 Ma, the rotation necessary to restore the Wyoming Province to the Superior craton fails to move the quartz diorite pole to the appropriate time-equivalent segment of the Superior APW path. PD = Ptarmigan dikes, MD = Matachewan dikes, MGD = Maguire dikes, SD = Senneterre dikes, BD = Biscotasing dikes, MND = Marathon dikes, MTD = Minto dikes, FFD = Fort Frances dikes; 1.500–1400 Ma Int. Mean = mean of 1500 to 1400 Ma intrusions from Laurentia; SFM = St. Francois Mountain volcanic pole; LAC/SG = mean pole from the Laramie anorthosite complex/Sherman Granite; ELG = Electra Lake Gabbro; and BSG = mean Belt Supergroup pole. Sources for the Superior paleomagnetic poles and ages are from Bates and Halls (1990), Buchan et al. (1993), Buchan et al. (1998), and the Online Global Paleomagnetic Database. Sources for the ca. 1500–1400 Ma poles from Laurentia are from Harlan and Geissman (1998) and Meert and Stuckey (2002).



44.3°N, 264.8°E ($\Omega = 172.6°$) in present-day southwestern Minnesota. Because the age of the quartz diorite is 2170 Ma, the most appropriate pole from the Superior Province for comparison with the Wind River paleomagnetic data is the Biscotasing dike pole of Buchan et al. (1993), which has a U–Pb age of 2167 Ma. Rotating the Wind River paleomagnetic poles (with the exception of the group B dikes that may be 1.4 Ga in age) using the Euler pole described above, the quartz diorite pole is shifted significantly away from the 2167 Ma part of the APW path (Fig. 8C), regardless of whether the in situ or tilt corrected poles are used. The pole does overlap the ca. 2.2 Ga part of the APW path, but this may be largely fortuitous, because the age of this part of the path is inconsistent with the geochronologic evidence of the quartz diorite and its magnetization reported here. One way to interpret the

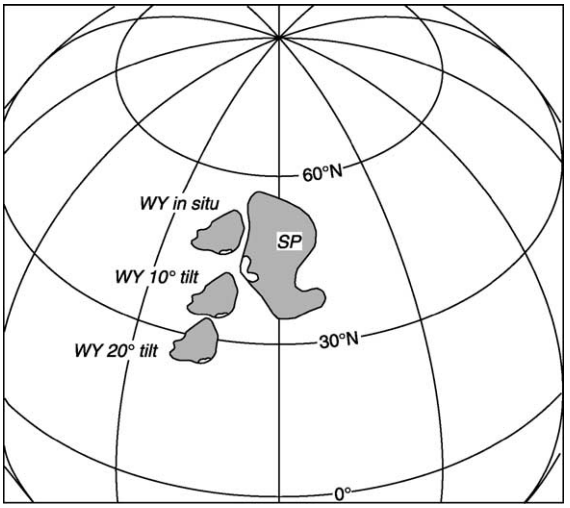


Fig. 9. Orthographic projection showing paleolatitudinal ca. 2170 Ma reconstruction of the Superior and Wyoming Provinces. The two cratons were rotated into their paleolatitudinal configurations using the Biscotasing dike pole of Buchan et al. (1993) for the Superior Province and by using the quartz diorite pole for the Wyoming Province. The cratons were then translated longitudinally to a closest approach configuration. Also shown is the possible range in paleolatitudes for the Wyoming Province based on possible tilts of the Wind River pole, as described in the text.

apparent discordance between the Wind River quartz diorite pole and the Biscotasing dikes pole is that the Wyoming Province and Superior cratons were in relative movement by about 2.1 Ga, following rifting of the Kenorland supercontinent as proposed by Roscoe and Card (1993). Alternatively, we note that the Proterozoic boundaries and margins of the two cratons are not well defined and the Euler pole of rotation we apply may be nothing but a crude approximation. Other possible configurations for the Wyoming and Superior provinces may be possible and need to be tested. Finally, we note that using a poorly defined Euler pole can yield misinterpretations regarding cratonic coherence.

Given the problems inherent with determining an appropriate Euler pole of rotation described above and the small paleomagnetic data base available for testing any reconstruction, an alternative method of testing the Roscoe and Card (1993) reconstruction involves rotating the Superior and Wyoming Provinces to their correct paleolatitudinal positions using the ca. 2170 Ma Biscotasing and quartz diorite poles and by then translating the two cratons longitudinally such that they attain a closest approach (cf. Meert and Stuckey, 2002). The Biscotasing and the in situ Wind River quartz diorite have essentially identical paleolatitudes (Biotasing dikes = 41.5° (+10.4°, −8.5°); quartz diorite = 46.5° (+6.6°, −5.8°)) and a fit based on minimizing paleolatitude differences indicates that the two cratons and the Huronian and Snowy Pass Supergroups could have been located adjacent to each other at about 2170 Ma (Fig. 9). This configuration is somewhat similar to that described by Roscoe and Card (1993) and is consistent with the hypothesis that the Huronian and Snowy Pass Supergroups were part of a single Early Proterozoic epicratonic basin. This interpretation is not significantly affected by modest tilts, especially given the uncertainties in whether paleosecular variation is averaged by the Wind River quartz diorite.

6. Conclusions

A large quartz diorite intrusion and crosscutting diabase dikes in the southern Wind River Range yield well-defined magnetizations that we interpret to be primary thermoremanent magnetizations. Isotopic dating of the quartz diorite indicates that it was emplaced at about 2170 Ma and is Early Proterozoic in age. $^{40}Ar/^{39}Ar$ age spectrum data indicate that the intrusion has not been significantly disturbed since emplacement. Previously, this dike was thought to be Archean in age and the 2170 Ma age appears to be unique with respect to known magmatic events in the Wyoming Province. Although the paleomagnetic data from eight sites may not have averaged paleosecular variation, the virtual geomagnetic poles from the quartz diorite and some diabase dikes are strongly discordant with respect to 2.4 to 2.0 Ga poles from the Superior Province. Application of any reasonable tilt corrections only increases the degree of discordance. Application of a rotation of the Wyoming Province and paleomagnetic poles using an Euler pole necessary to restore the two cratons following the proposed reconstruction of Roscoe and Card (1993) shows apparent discordance that could be interpreted to indicate that the two cratons were separated at about 2170 Ma. Reconstruction using a paleolatitudinal closest fit approach, however, suggests that the two cratons were

approximately adjacent at that time and that the Early Proterozoic Huronian and Snowy Pass Supergroups could have been part of single epicratonic sedimentary basin, as proposed by Roscoe and Card (1993). Given the possibility that these results may not average paleosecular variation and uncertainty regarding the age of the Wind River diabase dikes, we suggest that this conclusion must be viewed with some caution. Additional paleomagnetic data of Early Proterozoic age for the Wyoming Province are required in order to more robustly test the Roscoe and Card (1993) hypothesis, as well as other possible Wyoming and Superior Province configurations.

Acknowledgements

Chris Daniels and Arlo Weil provided valuable help in sample collection. The amphibole $^{40}Ar/^{39}Ar$ analyses were done by the lead author using facilities at the US Geological Survey in Denver, CO; R. Yeoman and L.W. Snee are acknowledged for their help in the isotopic dating process. A. Bekker, B.R. Frost, D. Evans, M.T.D. Wingate, and L.E. Apodaca are thanked for comments and constructive reviews of the manuscript. Bayley Harlan provided help with drafting. Orthographic reconstructions showing APW paths and plate reconstructions were made using programs written by C. McCabe and T. Torsvik. Financial support for part of this work was provided by NSF Grant EAR-8816814 and a Kelley-Silver Fellowship from the University of New Mexico.

References

Aleinikoff, J.N., Williams, I.S., Compston, W., Stuckless, J.S., Worl, R.G., 1989. Evidence for an Early Archean component in the Middle to Late Archean gneisses of the Wind River Range, west-central Wyoming: conventional and ion microprobe U–Pb data. Contributions to Mineralogy and Petrology 101, 198–206.

Bates, M.P., Halls, H.C., 1990. Regional variation in paleomagnetic polarity of the Matachewan dyke swarm related to the Kapuskasing Structural Zone, Ontario. Canadian Journal of Earth Sciences 27, 200–211.

Bayley, R.W., Proctor, P.D., Condie, K.C., 1973. Geology of the South Pass area, Fremont County, Wyoming. U.S. Geological Survey Professional Paper 793 (39 pp.).

Begemann, F., Ludwig, K.R., Lugmair, G.W., Min, K., Nyquist, L.E., Patchett, P.J., Renne, P.R., Shih, C.Y., Villa, I.M., Walker, R.J., 2001. Call for an improved set of decay constants for geochronological use. Geochimica et Cosmochimica Acta 65, 111–121.

Buchan, K.L., Mortensen, J.K., Card, K.D., 1993. Northeast-trending Early Proterozoic dykes of southern Superior Province: multiple episodes of emplacement recognized from integrated paleomagnetism and U–Pb geochronology. Canadian Journal of Earth Sciences 30, 1286–1296.

Buchan, K.L., Mortensen, J.K., Card, K.D., Percival, J.A., 1998. Paleomagnetism and U–Pb geochronology of diabase dyke swarms of Minto Block, Superior Province, Quebec, Canada. Canadian Journal of Earth Sciences 35, 1054–1069.

Chamberlain, K.R., Frost, B.R., 1995. Mid-Proterozoic mafic dikes in the central Wyoming Province: evidence for Belt-age extension and supercontinent breakup. Program with Abstracts—Geological Association of Canada–Mineralogical Association of Canada 20, 15.

Condie, K.C., Leach, A.P., Baadsgaard, H., 1969. Potassium–argon ages of Precambrian mafic dikes in Wyoming. Geological Society of America Bulletin 80, 899–906.

Dalziel, I.W.D., 1991. Pacific margins of Laurentia and East-Antarctica as a conjugate rift pair: evidence and implications for an Eocambrian supercontinent. Geology 19, 598–601.

Fisher, R.A., 1953. Dispersion on a sphere. Proceedings of the Royal Society of London, Series A 217, 295–305.

Frost, C.D., 1993. Nd isotopic evidence for the antiquity of the Wyoming Province. Geology 21, 351–354.

Frost, C.D., Frost, B.R., 1993. The Archean history of the Wyoming Province. In: Snoke, A.W., Steidtman, J.R., Roberts, J.R., Sheila, M. (Eds.), Geology of Wyoming. Geological Survey of Wyoming Memoir, vol. 5, pp. 58–76.

Frost, C.D., Frost, B.R., Chamberlain, K.R., Hulsebosch, T.P., 1998. The late Archean history of the Wyoming Province as recorded by granitic magmatism in the Wind River Range, Wyoming. Precambrian Research 89, 145–173.

Frost, B.R., Chamberlain, K.R., Swapp, S., Frost, C.D., Hulsebosch, T.P., 2000. Late Archean structural and metamorphic history of the Wind River Range: evidence for a long-lived active margin on the Archean Wyoming craton. Geological Society of America Bulletin 112, 564–578.

Hanley, L.M., Wingate, M.T.D., 2000. A SHRIMP zircon age for an Early Cambrian dolerite dyke—an intrusive phase of the Antrim Plateau Volcanics of northern Australia. Australian Journal of Earth Sciences 46, 1029–1040.

Harlan, S.S., Geissman, J.W., 1998. Paleomagnetism of the Middle Proterozoic Electra Lake Gabbro, Needle Mountains, southwestern Colorado. Journal of Geophysical Research 103, 15497–15507.

Harlan, S.S., Schmidt, C.J., Geissman, J.W., 1996. Nature and timing of recurrent movement on NW-trending faults in SW Montana: Middle Proterozoic to Neogene history. Abstracts with Programs-Geological Society of America 28 (7), A-509.

Harlan, S.S., Geissman, J.W., Snee, L.W., 1997. Paleomagnetic and $^{40}Ar/^{39}Ar$ geochronologic data from Late Proterozoic mafic dikes and sills, Montana and Wyoming. U.S. Geological Survey Professional Paper 1580 (16 pp.).

Hausel, W.D., 1988. Geologic map of the Radium Springs Quadrangle, including the Lewiston Gold District, Fremont County, Wyoming. Wyoming Geological Survey Map Series MS-26.

Hausel, W.D., 1991. Economic geology of the South Pass granite–greenstone belt, southern Wind River Range, western Wyoming. Report of Investigations-Geological Survey of Wyoming 44 (129 pp.).

Heaman, L.M., LeCheminant, A.N., 1993. Paragenesis and U–Pb systematics of baddeleyite (ZrO_2). Chemical Geology 110, 95–126.

Heaman, L.M., LeCheminant, A.N., Rainbird, R.H., 1992. Nature and timing of Franklin igneous events, Canada: implications for a Late Proterozoic mantle plume and the break-up of Laurentia. Earth and Planetary Science Letters 109, 117–131.

Kirschvink, J.L., 1980. The least-square line and plane and the analysis of paleomagnetic data. Geophysical Journal of the Royal Astronomical Society 62, 699–718.

Krogh, T.E., 1978. Vapor transfer for the dissolution of zircons in a multi-sample capsule at high-pressure. In: Zartman, R.E. (Ed.), Short Papers of the Fourth International Conference, Geochronology, Cosmochronology, Isotope Geology. U.S. Geological Survey Open-File Report OF78-701, pp. 233–234.

Krogh, T.E., Corfu, F., Davis, D.W., Dunning, G.R., Heaman, L.M., Kamo, S.L., Machado, N., 1987. Precise U–Pb isotopic ages of diabase dykes and mafic to ultramafic rocks using trace amounts of baddeleyite and zircon. Special Paper-Geological Association of Canada 34, 147–152.

Lanyon, R., Black, L.P., Seitz, H., 1993. U–Pb zircon dating of mafic dykes and its applications to the Proterozoic geological history of the Vestfold Hills, East Antarctica. Contributions to Mineralogy and Petrology 115, 184–203.

Love, J.D., Christiansen, A., 1985. Geologic Map of Wyoming. Scale 1:500,000.

Ludwig, K.L., 1980. Calculation of uncertainties of U–Pb data. Earth and Planetary Science Letters 46, 212–220.

Ludwig, K.L., 1985. PBDAT200: A Computer Program for processing raw Pb–U–Th isotope data. U.S. Geological Survey Open-File Report OF85-547. 54 pp.

McFadden, P.L., McElhinny, M.W., 1988. The combined analysis of remagnetization circles and direct observations in paleomagnetism. Earth and Planetary Science Letters 87, 161–172.

Meert, J.G., Stuckey, W., 2002. Revisiting the paleomagnetism of the 1.476 Ga St. Francois Mountains igneous province. Tectonics 21 (2), 1–19.

Min, K., Mundil, R., Renne, P.R., Ludwig, K.R., 1999. A test for systematic errors in $^{40}Ar/^{39}Ar$ geochronology through comparison with U/Pb analysis of a 1.1-Ga rhyolite. Geochimica et Cosmochimica Acta 64, 73–98.

Moores, E.M., 1991. Southwest-U.S.–East Antarctic (SWEAT) connection: a hypothesis. Geology 19, 425–428.

Premo, W.R., Van Schmus, W.R., 1989. Zircon geochronology of Precambrian rocks in southeastern Wyoming and northern Colorado. In: Grambling, J.A., Tewksbury, B.J. (Eds.), Proterozoic Geology of the Southern Rocky Mountains. Geological Society of America Special Paper 235, pp. 13–32.

Premo, W.R., Helz, R.T., Zientek, M.L., Langston, R.B., 1990. U–Pb and Sm–Nd ages for the Stillwater Complex and its associated sills and dikes, Beartooth Mountains, Montana: identification of a parent magma? Geology 18, 1065–1068.

Renne, P.R., Deino, A.L., Walter, R.C., Turrin, B.D., Swisher III, C.C., Becker, T.A., Curtis, G.H., Sharp, W.D., Jaouni, A.-R., 1994. Intercalibration of astronomical and radioisotopic time. Geology 9, 783–786.

Romer, R.L., Scharer, U., Wardle, R.J., Wilton, D.H.C., 1995. U–Pb age of the Seal lake Group Labrador: relationship to Mesoproterozoic extension-related magmatism of Laurasia. Canadian Journal of Earth Sciences 32, 1401–1410.

Roscoe, S.M., Card, K.D., 1993. The reappearance of the Huronian in Wyoming: rifting and drifting of ancient continents. Canadian Journal of Earth Sciences 30, 2475–2480.

Royse Jr., F., 1993. An overview of the geologic structure of the thrust belt in Wyoming, northern Utah, and eastern Idaho. In: Snoke, A.W., Steidtmann, J.R., Roberts, S.M. (Eds.), Geology of Wyoming. Geological Survey of Wyoming Memoir, vol. 5, pp. 272–311.

Samson, S.D., Alexander Jr., E.C., 1987. Calibration of the interlaboratory $^{40}Ar–^{39}Ar$ dating standard MMhb-1. Chemical Geology 66, 27–34.

Spall, H., 1971. Paleomagnetism and K–Ar age of mafic dikes from the Wind River Range, Wyoming. Geological Society of America Bulletin 82, 2457–2472.

Stacey, J.S., Kramers, J.D., 1975. Approximation of terrestrial lead isotope evolution by a two stage model. Earth and Planetary Science Letters 26, 207–221.

Steiger, R.H., Jäger, E., 1977. Subcommission on geochronology: convention on the use of decay constants in geo- and cosmochronology. Earth and Planetary Science Letters 36, 359–362.

Stuckless, J.S., Hedge, C.E., Worl, R.G., Simmons, K.R., Nkomo, I.T., Wenner, D.B., 1985. Isotopic studies of the Late Archean rocks of the Wind River Range, Wyoming. Geological Society of America Bulletin 96, 850–860.

Van der Voo, R., Grubbs, K.L., 1977. Paleomagnetism of the Triassic Chugwater redbeds revisited, Wyoming, U.S.A. Tectonophysics 41, 27–33.

Williams, H., Hoffman, P.F., Lewry, J.F., Monger, J.W.H., Rivers, T., 1991. Anatomy of North America: thematic geologic portrayals of the continent. Tectonophysics 187, 117–134.

Wingate, M.T.D., Giddings, J.W., 2000. Age and palaeomagnetism of the Mundine Well dyke swarm, Western Australia: implications for an Australia–Laurentia connection at 755 Ma. Precambrian Research 100, 335–357.

York, D., 1969. Least-squares fitting of a straight line with correlated errors. Earth and Planetary Science Letters 5, 320–324.